CS19541 – COMPUTER NETWORKS LAB

PRACTICAL – 8

NMAP LIVE HOST DISCOVERY

Aim

To identify active hosts on a network using Nmap discovery methods, including ARP, ICMP, TCP and UDP.

Objective

To learn how Nmap detects systems that are online before carrying out additional network scans.

Software Required

• Nmap
• TryHackMe Nmap room / AttackBox
• Computer with network access

Introduction

Nmap (Network Mapper) is a free, open-source utility used to locate active devices and learn about network services. In this practical, ARP, ICMP, TCP and UDP probes are considered as different ways to discover hosts.

Host discovery is generally performed before more detailed scanning. ARP is useful for targets on a local Ethernet network, while ICMP, TCP and UDP probes can help determine whether a target is responding.

1. ARP Host Discovery

Address Resolution Protocol (ARP) is used on a local Ethernet network to find the hardware/MAC address associated with an IP address. An ARP reply indicates that the target can be reached on the local network.

Command:

```
nmap -PR -sn TARGETS
```

ARP discovery is limited to the local subnet because ARP operates at the link layer.

2. ICMP Host Discovery

Nmap can transmit an ICMP Echo Request (Type 8) and check for an Echo Reply (Type 0). Receiving a reply shows that the target is responding.

Command:

```
nmap -PE -sn TARGETS
```

Other ICMP discovery options include -PP for ICMP Timestamp requests and -PM for ICMP Address Mask requests.

3. TCP SYN Host Discovery

TCP SYN Ping sends a TCP packet with the SYN flag to a selected port. A response such as SYN/ACK or RST can indicate that the host is active.

Command:

```
nmap -PS23 TARGETS
```

In the lab context, TCP SYN Ping is described as not requiring a privileged account, whereas TCP ACK Ping requires a privileged account.

4. UDP Host Discovery

UDP Ping sends a UDP probe to a chosen port. If that port is closed, the target may return an ICMP Port Unreachable message, which can indicate that the host is active.

Command:

```
nmap -PU40125 TARGETS
```

Important Nmap Commands

Purpose	Command / Option
Host discovery only	nmap -sn TARGETS
ARP discovery	nmap -PR -sn TARGETS
ICMP Echo	nmap -PE -sn TARGETS
ICMP Timestamp	nmap -PP -sn TARGETS
ICMP Address Mask	nmap -PM -sn TARGETS
TCP SYN Ping	nmap -PS23 TARGETS
TCP ACK Ping	nmap -PA80 TARGETS
UDP Ping	nmap -PU40125 TARGETS
Disable reverse DNS	nmap -n TARGETS
Reverse DNS lookup	nmap -R TARGETS

Observation

1. ARP can identify hosts on the same local subnet.
2. ICMP Echo checks whether a host replies to an Echo Request.
3. TCP SYN Ping sends a SYN probe to a selected TCP port.
4. TCP ACK Ping uses an ACK probe; in the lab context, it is described as requiring a privileged account.
5. UDP Ping may reveal an active host through an ICMP Port Unreachable response from a closed UDP port.
6. The -sn option performs host discovery without continuing to a port scan.

Observation Table

Scan Type	Option	Expected Observation
ARP	-PR	ARP reply / MAC information from a local host
ICMP Echo	-PE	ICMP Echo Reply
ICMP Timestamp	-PP	ICMP Timestamp Reply
ICMP Address Mask	-PM	ICMP Address Mask Reply
TCP SYN	-PS	SYN/ACK or RST response
TCP ACK	-PA	An RST response can indicate a live host
UDP	-PU	ICMP Port Unreachable may indicate a live host

Result

Thus, Nmap live host discovery was studied using ARP, ICMP, TCP and UDP techniques.

Conclusion

Nmap offers several methods for locating active systems on a network. ARP is mainly applicable to the local Ethernet network, while ICMP, TCP and UDP probes provide other ways to check whether a target host is responding.